

The 67th ASH Annual Meeting Abstracts

POSTER

722. ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD AND IMMUNE RECONSTITUTION

VIP pathway inhibition alters microbial metabolism to balance graft-versus-leukemia and gvhd

Kayla Robinson¹, Shayna Jankowski¹, Jian-Ming Li¹, Edmund Waller¹¹ Emory University, Neuroscience, Atlanta, United States

Abstract Introduction: Emerging evidence highlights the gut microbiome as a critical player in gastrointestinal acute graft-versus-host disease (GvHD): microbial diversity, short-chain fatty acid (SCFA) production, and epithelial integrity all influence the incidence of acute GvHD, immune recovery and GvHD outcomes.

Vasoactive intestinal peptide (VIP) is an immunosuppressive neuropeptide involved in maintaining homeostasis in the gut microbiome. VIP knockout (VIP-KO) mice exhibit greater anti-tumor immunity compared to wild-type, but have severe GvHD after allogeneic bone marrow transplant (allo-BMT) with significantly reduced survival. Furthermore, VIP-KO mice have a distinct gut microbiome that has been associated with resistance to the growth of transplantable leukemia cell lines. ANT308 is a VIP-receptor antagonist designed to block signaling through VIP-receptors VPAC1 and VPAC2. We hypothesized that pharmacological inhibition of VPAC signaling would recapitulate some of the effects of VIP-KO on the gut microbiome. We compared the effects of ANT308, a high-affinity VPAC antagonist, on the murine gut microbiome and the incidence and severity of GvHD to VIP-KO recipients in allogeneic bone marrow transplant (allo-BMT) models.

Methods: Wild-type C57Bl/6 mice and VIP-KO mice were cohoused to exchange microbiota via coprophagy for 4–6 weeks. In another group, wild-type C57Bl/6 mice were treated with daily subcutaneous injections of 20ug ANT308 for 4 weeks. Both groups received allo-BMT from B10.BR donor mice. Mouse weight, GvHD score, and survival were monitored. Stool samples were collected pre- and post- VIP-KO cohousing, pre- and post- ANT308 treatment, and post-transplant for shotgun sequencing and untargeted lipidomic analysis. Taxonomic and functional data were analyzed using MaAsLin2 analysis. MetaCyc pathway mapping identified metabolic and biosynthetic pathways involved in SCFA and lipid synthesis. Lipidomic profiling of fecal pellets employed untargeted mass spectrometry. Non-parametric tests were used for differential abundance analysis, and pathway shifts were visualized via heatmaps and volcano plots.

In a graft-versus-leukemia model (GvL) of allo-BMT, wild-type mice cohoused with VIP-KO mice for 4–6 weeks were inoculated with C1498-luciferase and received allo-BMT from B10.BR donors. Recipient mice were monitored for weight, GvHD score, survival, and C1498-luc growth was measured by IVIS imaging.

Results: Transfer of the VIP-KO microbiome to wild-type mice following cohousing with VIP KO mice was confirmed by taxonomic and functional composition from shotgun sequencing of stool samples. Pharmacological inhibition of the VIP-signaling pathway with ANT308 treatment revealed a microbiome similar to VIP-KO mice, with increased abundance of short-chain fatty acids (SCFA)-producing taxa, including *Muribaculum intestinale* and *Lactobacillus murinus*. Metabolic pathways supporting pyruvate fermentation to acetate & lactate I/II pathways were enriched and correlated with known SCFA producer taxa in ANT308-treated wild-type mice and wild-type mice with VIP-KO microbiome. Lipidomic analysis revealed enrichment of LPC 18:2 in ANT308 and VIP-KO microbiome, and apigenin-6-C-glucoside-8-C-arabinoside in ANT308-treated microbiome only.

Wild-type mice with the adoptive transfer of VIP-KO microbiome had improved median survival time and showed greater leukemia (C1498-luc) control in a GvL model compared to wild-type mice, but had worse GvHD scores with more weight loss. Interestingly, wild-type mice treated pre-transplant with ANT308 reconstituted with a microbiome characterized by increased SCFA synthesis and had improved GvHD scores, less weight loss, and greater survival compared to untreated wild-type mice. Survival benefit of ANT308-reconstituted microbiome in a

Conclusion: Modulation of VIP signaling represents a promising strategy to shape gut microbial metabolism in the context of allo-BMT. Our findings suggest that microbiome-derived metabolites enriched through VIP pathway inhibition may contribute to improved immune balance and transplant outcomes. Future studies will focus on defining the mechanistic roles of these

metabolites and evaluating their potential as therapeutic targets to enhance graft-versus-leukemia effects while mitigating GvHD.

<https://doi.org/10.1182/blood-2025-6034>